

Latent FMCG Demand Ceiling Estimation & Potential-Based Trade Allocation

Data Storm 7.0 - Round 2 Comprehensive Methodology and Technical Paper

Team	Stack Kings
Competition	Data Storm 7.0 - 2nd Round
Scope	20,000 Outlets · 10 Distributors · 4 Provinces · Sri Lanka
Period	January 2026 Demand Forecast

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Abstract This paper presents the complete technical methodology employed by Team Stack Kings in the Data Storm 7.0 Round 2 competition. We detail a five-phase, end-to-end analytics pipeline that spans raw data ingestion through a Medallion architecture, rigorous data quality enforcement, external Points of Interest (POI) acquisition via OpenStreetMap, gravity-model spatial feature engineering, right-censored latent demand ceiling estimation, an LP-based promotional spend optimizer, and a transparent Generative AI explainability layer. The pipeline produces 20,000 outlet-level FMCG demand ceiling predictions and a LKR 5,000,000 optimized trade-spend allocation across 9,000 Western Province outlets, achieving an estimated 1,004,555 incremental litres (+253% vs. naive equal-split baseline) and passing all 327 automated audit checks.

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1. DATA ENGINEERING AND POI PIPELINE

1.1 Medallion Architecture

The ingestion and transformation pipeline follows a four-layer Medallion architecture, ensuring full lineage, reproducibility, and quarantine transparency. Each layer is idempotent: re-running the pipeline from the Silver layer will fully regenerate Gold and Prediction outputs without duplicating data.

Layer	Role	Location
Bronze	Immutable ingest; SHA-256 manifest for tamper detection	bronze/raw/
Silver	DE checks, cleaning, quarantine with failure_reason codes	silver/clean/ · silver/quarantine/
Gold	POI + spatial features + statistical aggregates	gold/features/
Predictions	Ensemble model + LP optimizer outputs	gold/predictions/

Table 1.1 — Medallion layer definitions and data flow.

Note: 37,205 records were quarantined with explicit `failure_reason` codes no records are silently discarded.

1.2 Reusable Data Quality Checks

Five parameterised data quality functions are implemented in `src/de_checks.py` and applied consistently across all datasets via `phase2_silver.py`:

Check	Description
Duplicates	Primary-key deduplication across transaction records
Nulls	Mandatory field null detection with configurable thresholds
Referential Integrity	Foreign key validation across outlet, distributor, product tables
Value Range	Domain-specific min/max enforcement (e.g., volume ≥ 0)
Format / Type	Date format, numeric type, and string pattern validation

Table 1.2 — Data quality check catalogue.

1.3 External POI Acquisition

Points of Interest (POIs) form the spatial backbone of the footfall proxy features. Production acquisition queries the **OpenStreetMap Overpass API** with a single Sri Lanka bounding box (5.9°N–9.9°N, 79.6°E–81.9°E) across seven categories: education, health, transport, market, tourism, food, and worship. API responses are cached under `gold/features/poi_raw/` to avoid redundant requests.

Fallback strategy: When the Overpass API returned HTTP 406 in the build environment, `phase3_poi_synthetic.py` generated geographically calibrated synthetic POIs aligned to published OSM node statistics. Re-running `phase3_poi_acquire.py` with internet connectivity automatically replaces synthetic data.

1.4 Footfall and Market-Potential Proxies

Spatial signals are engineered per outlet after geocoding (lat/lon repair includes swap correction and zero-coordinate imputation). Two generations of features are maintained:

Feature Generation	Description	Use
Legacy (A/B)	count_<cat>_1km / 3km · nearest_<cat>_m (disk counts)	Round 1 baseline
Round 2 (Production)	Exponential decay influence $\sum \exp(-\beta \cdot d)$ per category	Final predictions
Competition	Outlet-to-outlet density · DBSCAN cluster zones · saturation labels	Ensemble adjustment

Table 1.4 — Feature generation hierarchy 1

1.5 Distance-Decay β Calibration and Sensitivity

The core spatial model computes per-outlet influence from each POI category using an exponential distance-decay kernel. For outlet i and POI j in category c :

$$\text{decay}_{ic} = \sum_j \exp(-\beta_c \cdot d_{ij}) \text{ where } d_{ij} \text{ is in km; POIs beyond 10 km excluded}$$

β values are hand-calibrated with domain priors rather than grid-searched on unobserved January 2026 labels, implementing gravity-style decay consistent with the competition PDF:

Tier	β_c (1/km)	Categories	Half-distance $d_{1/2} = \ln(2)/\beta$
Fast	3.0	transport, food	~231 m
Medium	2.0	health, market, tourism	~347 m
Slow	1.5	education, worship	~462 m

Table 1.5a — Production β calibration by POI category tier.

Rationale: trip-chaining and food purchases are hyper-local (high β); schools and temples anchor wider community traffic (low β); health, market, and tourism sit between. Default for unknown categories: $\beta = 2.0$.

Single-POI decay weights $\exp(-\beta \cdot d)$:

d (km)	$\beta = 1.5$	$\beta = 2.0$	$\beta = 3.0$
0.25	0.69	0.61	0.47
0.50	0.47	0.37	0.22
1.00	0.22	0.14	0.05
2.00	0.05	0.02	0.002

Table 1.5b — Decay weights at key distances. At $\beta=3.0$ a POI at 5 km contributes <0.1% of one at origin; the 10 km cap is computational, not material for fast-decay categories.

Deployed scale ($n = 19,960$ geocoded outlets):

Feature	% Non-zero	Median	p90
decay_worship	100%	1.35	2.82
decay_education	100%	0.63	1.52
decay_transport	98.8%	0.06	0.40
decay_total	100%	2.51	4.85

Table 1.5c — Feature coverage statistics from metadata/gold_spatial_report.csv.

Sensitivity vs. legacy 3 km disk counts (Spearman ρ):

Category	β	Spearman ρ	Notes
worship	1.5	0.88	High rank agreement
tourism	2.0	0.87	High rank agreement
education	1.5	0.84	High rank agreement
market	2.0	0.81	Moderate–high
health	2.0	0.79	Moderate–high
food	3.0	0.66	Lower - fast decay diverges from flat ring
transport	3.0	0.61	Lowest - fast decay diverges most

Table 1.5d — ρ is lower for fast-decay categories: decay is not a monotone rescaling of flat disk counts; this is the intended move to gravity-style proximity weighting.

Reproducibility: `python src/phase3_gold_features.py → A/B exports; python src/summarize_decay_beta.py → metadata/decay_beta_sensitivity.csv`

2. DATA CLEANING AND ARTIFACT NEUTRALISATION

All data artifacts are classified, counted, and treated systematically. Forensic analysis is performed by `src/phase1_forensics.py`, producing `metadata/forensics_report.csv`. Silver-layer audit results are exported to `metadata/silver_audit.csv`.

Artifact	Count	Treatment Strategy
Duplicate transaction PKs	32,240	Quarantine with <code>failure_reason = 'duplicate_pk'</code>
Negative volumes	4,753	Quarantine physically impossible observations
Blackout outlets (no Dec 2025)	7,417	Retained interpreted as supply signal, not zero demand
Swapped lat/lon coordinates	200	Auto-corrected via country bounding box heuristic
Zero coordinates	40	Flagged; province-median spatial imputation applied
Outlet type typos	785	Normalised (e.g., 'Grocry' → 'Grocery')

Table 2.1 — Artifact inventory and treatment. Total quarantined: 37,205 records.

Key design decision — Blackout outlets: The 7,417 outlets with no December 2025 transactions are *retained* rather than dropped. The absence of sales is treated as a supply-side or distribution constraint signal, not evidence of zero demand. This choice is critical for the ceiling estimation task: dropping these outlets would introduce selection bias toward well-served locations.

3. MATHEMATICAL FRAMEWORK

3.1 Right-Censored Demand (Not Left-Censored)

The fundamental statistical challenge is that observed monthly sales volume is a **lower bound** on true latent demand — not a direct measurement. Supply constraints, credit limits, and delivery restrictions cap what an outlet can purchase regardless of actual demand:

```
Y_obs = min(Y*, C) where C = supply / credit / delivery constraint
Therefore: E[Y_obs] < E[Y*] (observed mean understates latent demand)
```

The modelling target is the **Maximum Monthly Purchase Potential** — the upper tail of the latent demand distribution Y^* — not the censored conditional mean. This framing shapes all downstream modelling choices.

3.2 Why Not Tobit?

Tobit regression is the canonical model for censored outcomes. However, Tobit MLE requires explicit censoring indicators $\delta_i \in \{0, 1\}$ identifying which observations were censored. This dataset provides no δ_i — every sale may or may not have hit a supply cap. Without δ_i the Tobit likelihood is unidentified. We document this gap explicitly rather than misapply the model and introduce false confidence in parameter estimates.

3.3 Latent Ceiling Methods

Three complementary methods estimate the demand ceiling, then are combined via an ensemble:

Method A — Lookalike Cluster Ceilings ($K = 50$)

K-Means clustering ($K=50$) is applied on a feature set combining historical volume, outlet structural attributes, and production decay POI features. For each cluster, the ceiling is defined as the 90th percentile of peer outlets' `max_monthly_vol`. This leverages cross-outlet peer comparison: if similar outlets have sustained higher volumes, the target outlet's constraint is likely supply-side rather than demand-side.

Method B — Quantile Regression ($\tau = 0.90$)

A quantile regression model at $\tau = 0.90$ is trained on decay POI features and outlet structural covariates, targeting the 90th percentile of observed volume. The pinball loss function ensures the model optimises for the upper tail rather than the mean, producing a conservative upper-bound estimate that reflects spatial and structural drivers.

Method C — Ensemble + Competition Adjustment

The two ceiling estimates are combined and adjusted for local competitive saturation:

```
base_ensemble = max(kmeans_ceiling, qr_ceiling)
adjusted = base * (1 - 0.20 * norm_density) * (1 + 0.10 * (1 - norm_density))
prediction = max(adjusted * jan_factor, own_max_vol, 1.0)
```

A soft floor anchored to cooler capacity is also applied: when an outlet has coolers, the prediction is floored at 80% of replenishment capacity (`cooler_count × 50 L × 4 cycles/month`), acting as a supply-cycle proxy.

Validation Metric	Result
Total predictions generated	20,000 (100% of outlets)
Predictions below own historical max	0 (hard constraint enforced)
Median uplift over historical max	1.26× (26% demand ceiling above observed peak)
Backtest coverage (Dec 2025 data)	100%

Table 3.3 — Ensemble validation metric

4. SPEND OPTIMISATION LOGIC (LKR 5M · WESTERN PROVINCE)

The LP-based spend optimizer allocates a LKR 5,000,000 promotional budget across 9,000 Western Province outlets served by distributors DIST_W_01, DIST_W_02, and DIST_W_03, targeting January 2026 demand uplift.

Objective Function

The optimiser maximises **incremental volume vs. the 3-month recent baseline** (not raw potential ranking, which would concentrate spend on already-high-volume outlets with smaller marginal returns):

$$\Delta \text{vol}(\mathbf{x}) = \text{gap} \times (1 - \exp(-\alpha \cdot \mathbf{x} / 1000))$$

$\text{gap} = \max(\text{predicted_ceiling} - 3\text{-month_baseline}, 0) \mid \alpha \text{ density-adjusted per outlet}$

Solver

The non-linear response function is linearised into 3–5 piecewise-linear spend segments per outlet (0 → 500 → 2K → 10K → 50K LKR), enabling solution via `scipy.optimize.linprog` with the HiGHS backend — a highly scalable LP solver suitable for the 9,000-outlet problem size.

Constraints

Constraint	Specification
Total budget	$\sum x_i \leq \text{LKR } 5,000,000$
Non-negativity	$x_i \geq 0$ for all outlets
Per-outlet cap	$\min(\text{LKR } 50,000, 5\% \times \text{gap} \times \text{unit_price})$
Min spend floor	LKR 2,000 on top-100 gap outlets (when LP-feasible)

Table 4.1 — LP constraint set.

Optimisation Results

Metric	Value
Total spend deployed	LKR 5,000,000 (100% utilised)
Estimated incremental volume	~1,004,555 L
Return on investment	~201 L / LKR 1,000 spent
Lift vs. naïve top-500 equal split	+253%
DIST_W_01 allocation	LKR 1,670,000
DIST_W_02 allocation	LKR 1,660,000
DIST_W_03 allocation	LKR 1,680,000

Table 4.2 — Optimisation results from metadata/optimization_report.csv.

Distributor balance: Spend is distributed near-evenly across the three Western Province distributors (~LKR 1.67M each), reflecting comparable outlet density and gap profiles rather than a concentration artefact. Outputs:

`submissions/StackKings_budget_allocations.csv` (9,000 rows) and
`gold/predictions/budget_allocations_raw.csv`.

5. GENERATIVE AI TRANSPARENCY LOG

5.1 Tools and Roles

Tool	Role in Pipeline
Antigravity / Gemini (IDE)	Pipeline design, Python/TypeScript code generation, debugging, audit scaffolding
Ollama gemma4:e4b (local)	Optional XAI narration in web app (primary LLM)
Gemini 2.0 Flash (API)	Optional cloud XAI fallback when Ollama unavailable

Table 5.1 — GenAI tool inventory.

Critical constraint: GenAI does not generate predictions. All litre forecasts and spend allocations are produced by deterministic Python pipelines. LLMs are used exclusively to narrate pre-computed outlet JSON for the explainability (XAI) layer.

5.2 Representative Development Prompts

The following human directives drove the iterative codebase construction:

- [Phase 3 POI] *"Proceed to phase 3 – acquire POI via Overpass with caching and silver/gold integration."*
- [Decay features] *"Implement exponential decay POI and competitor density per final-round PDF."*
- [Ensemble] *"Wire K-Means + QR ensemble into submissions; validate ≥ own max."*
- [Optimizer] *"Build LKR 5M Western optimizer with diminishing returns and audit checks."*
- [Web app] *"Scaffold Next.js app with hybrid Ollama → Gemini → template XAI."*

After each phase: `python src/audit_all.py` (current state: **327 PASS / 0 FAIL**).

5.3 In-App XAI Prompts (Exact)

Svstem prompt (Ollama):

You are a FMCG analytics assistant. Explain predictions using only provided data. Write exactly 3 short paragraphs in plain business language.

User prompt (Ollama + Gemini — buildXaiPrompt):

Explain this FMCG outlet prediction in 3 short business paragraphs. Use ONLY the facts in the JSON below. Include feature importance from `modelDrivers.qrTopDrivers` (weight and contributionLiters) and competition adjustment weights. `{buildXaiPayload: outlet summary + modelDrivers.qrTopDrivers + modelDrivers.competition + ...}`

Resolution order:

Ollama (think: false · T=0.2 · num_predict=512) → Gemini 2.0 Flash (T=0.2 · max 300 tokens) → `buildTemplateExplanation()` rules

5.4 Validation and Human Review

Step	Method
Automated	python src/validate_xai_samples.py — 20/20 template + QR weights + competition terms
Live LLM spot-check	python src/validate_xai_llm.py — optional Ollama/Gemini (skips if offline)
Schema	export_manifest.json schema v2 with modelDrivers field
Human review	Team spot-checked Western high-gap outlets: LLM/template outputs matched JSON numbers
Demo resilience	Template fallback always available; app functions without LLM connectivity

Table 5.4 — XAI validation steps. 1

Full disclosure: genai_transparency_log.md

6. DELIVERABLES AND REPRODUCIBILITY

All competition deliverables are self-contained and reproducible from a single pipeline command. The full codebase, outputs, and web application are structured as follows:

Deliverable	Path / Command	Description
Predictions CSV	submissions/StackKings_predictions.csv	20,000 rows — one per outlet
Budget CSV	submissions/StackKings_budget_allocations.csv	9,000 Western Province outlets
Source code	src/ · README.md	Full Python pipeline
Web application	app/	Tailwind UI: npm run build:clean && npm run start
One-command pipeline	python src/run_round2_pipeline.py	End-to-end Bronze→Predictions
Pre-submit QA	python src/verify_all.py	Final validation gate

Idempotency guarantee: Re-running `run_round2_pipeline.py` from Silver inputs will fully regenerate Gold features, model predictions, and optimizer outputs without data duplication or state corruption.

7. KEY RESULTS SUMMARY

The following table consolidates the headline quantitative outcomes across all pipeline phases:

Category	Metric	Value
Data Engineering	Records quarantined (with failure codes)	37,205
Data Engineering	Geocoded outlets (non-zero decay_transport)	19,960 (98.8%)
Data Engineering	Pipeline audit checks	327 PASS / 0 FAIL

Data Cleaning	Duplicate transaction PKs	32,240 quarantined
Data Cleaning	Negative volume records	4,753 quarantined
Data Cleaning	Blackout outlets retained	7,417
Spatial Features	POI categories	7 (education, health, market, transport, tourism, food, worship)
Spatial Features	Max POI search radius	10 km
Spatial Features	Spearman ρ range (decay vs flat counts)	0.61 – 0.88
Demand Modelling	Outlets predicted	20,000 (100%)
Demand Modelling	Predictions below own max	0
Demand Modelling	Median demand uplift	1.26×
Spend Optimisation	Budget allocated	LKR 5,000,000 (100%)
Spend Optimisation	Estimated incremental volume	~1,004,555 L
Spend Optimisation	ROI	~201 L / LKR 1,000
Spend Optimisation	Lift vs. naïve equal split	+253%

Table 7.1 — Consolidated results dashboard.